

Lightweight Machine Learning for Detecting DGA-Based (Domain Generation Algorithm Based) Botnet Communications

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Sean Silver Ethan Ge
seanrs@ucla.edu ethange@ucla.edu

Lihao Pan
lihaopan@g.ucla.edu

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Sean Silver

University of California, Los Angeles

Ethan Ge

University of California, Los Angeles

Lihao Pan

University of California, Los Angeles

Abstract

Domain Generation Algorithms (DGAs) are used by malware and botnets to evade static blocklists by generating disposable command-and-control domains. This paper examines whether DGA-generated domains can be detected using a lightweight Machine Learning approach based only on lexical features from domain strings, without relying on DNS histories, WHOIS records, malware analysis, or deep learning. Using malicious domains from an open-source DGA dataset and benign domains from the Cisco Umbrella Top 1 Million list, we train and evaluate a Random Forest classifier across balanced and class-imbalanced settings. Features include domain length, entropy, vowel and digit ratios, unique-character count, hyphen count, consecutive consonant and digit patterns, subdomain count, suspicious top-level-domain indicators, Punycode usage, and character-class entropy. Results show near-perfect performance when structural indicators are included, though ablation experiments suggest possible dataset-specific bias. After removing potentially leakage-prone features, the model still achieves 92.4% accuracy, 92.5% F1 score, and 0.984 ROC-AUC. Overall, the findings suggest that lightweight, lexical models can serve as efficient, interpretable first-stage detectors for DGA-based botnet communication but further validation on live DNS traffic and diverse DGA families is necessary.

Keywords

DGA, Benign Domain, Malicious Domain, Lexical Feature, Botnet

1 Introduction

Malicious network activity is an ever growing concern for cybersecurity defenses to contain. This unauthorized behavior can lead to data breaches, malware distribution, and service interruptions. Malicious actors will abuse the Domain Name System (DNS) to register domains for their distributions as

this traffic is easy to manipulate and a core component to Internet communications. Thus, domain-based infrastructure has become a major concern in the cybersecurity community. Botnets, for example, rely on command-and-control communication so that infected machines can receive instructions from an attacker. When this communication uses predictable domains or IP addresses, defenders can block those locations directly. However, many modern botnets use more flexible techniques to hide or constantly change their command-and-control infrastructure, making detection a harder problem for traditional security systems.

A specific version of this problem is detecting malicious domains produced by Domain Generation Algorithms, or DGAs. Instead of using one fixed domain, DGA-based malware can generate many possible domain names and attempt to contact them until one connects to the attacker's infrastructure. These generated domains often look different from normal domains because they may be longer, more random, less pronounceable, or have unusual character patterns and lack histories that would be indicative of a commonly used and reputable domain.

While complex deep learning models and resource-heavy DNS traffic analysis offer high detection rates, their computational overhead limits real-time deployment at the network edge. Previous work in DGA detection has neglected a minimal, interpretable, lightweight, lexical feature framework that could be applied to real-world DNS traffic. This study addresses this limitation by evaluating a lightweight, purely lexical machine learning approach. Specifically, we structure our investigation around three core research questions: **Q1: Can a lightweight, lexical machine learning model effectively isolate DGA domains from benign ones?** Our evaluation demonstrates that a highly optimized Random Forest model achieves an exceptional $\sim 99.9\%$ accuracy and F1-score with negligible computational and memory overhead. **Q2: What specific lexical features serve as the most critical signals for defining DGA behavior?** Our analysis reveals a heavily skewed feature dependency; structural features, specifically subdomain count and malicious top-level domains, can drive

over 90% of the model’s classification decisions. Withholding these structural features reveals that consecutive consonants, domain length, and hyphen count can also drive over 90% of the model’s classification decisions with a notable, but not overly detrimental, impact on accuracy. **Q3: Is a lightweight lexical model practically deployable based on comprehensive real-world performance metrics?** By evaluating accuracy, precision, recall, F1 score, and false-positive rates alongside a rigorous ablation study, we demonstrate that our model functions optimally as a high-speed, first-stage triage filter rather than a standalone shield.

By explicitly defining these structural components, this paper provides a transparent narrative mapping lightweight feature engineering directly to robust, deployable botnet detection. Our research contributes to DGA detection by analyzing a unique curated lexical feature set, revealing important features through an ablation study, and classifying a Random Forest model that can be utilized as a practical first-stage triage filter. With this low-compute model, we can flag suspicious domains for further inspection at DNS resolvers.

2 Related Work

Where Related Work Should Go

We include the related work section after the introduction section as our paper requires an explanation of previous studies to understand our intentions. We are in a crowded field where other researchers have developed models similar to a lightweight, lexical DGA detection model, and many may question the simplicity/feasibility. Therefore, it is essential to our methodology that we compare our model to others. We need to know if our model is actually successful by comparing it to more heavyweight models or other lexically-defined models. We would build off of them to ensure that our model’s efficacy is tangible and must explain why our model stands out among other options, whether through performance metrics or unique feature analysis.

Prior work on DGA detection includes practical systems that identify botnet behavior in real network environments. Antonakakis et al. [1] introduced Pleiades, a DNS-based system that detects DGA-based malware by analyzing failed DNS lookups or NXDOMAIN traffic, showing that DGA detection can operate at large scale and can uncover botnet activity in real-world network settings. Pleiades, however, relies primarily on DNS traffic behavior rather than only the domain string itself which would be the more lightweight and restricted categorization of DGA-generated domains purely using lexical features.

Other studies have applied deep learning to detect DGA domains and classify botnet families. Ravi et. al. [3] propose an integrated deep learning framework for detecting DGA botnets and DNS homograph attacks, while also evaluating the robustness of the model against adversarial botnet evasion

techniques such as DeepDGA and CharBot. Tuan et al. [5] also uses advanced neural architectures, including LSTM networks and attention mechanisms, to detect and classify DGA botnet families. These existing works provide strong baselines for DGA detection via deep learning but not with lightweight machine learning models trained only on lightweight lexical indicators.

Sun and Liu [4] provide a highly useful comparison point, as their work combines multiple feature types and reports standard evaluation metrics on models, utilizing domain name, WHOIS, and n-gram features for a deep learning based detection strategy. However, it differs from our approach because it is not limited to lexical features alone. By comparing the accuracy, precision, recall, and F1 score against a rich feature-based model, we can better evaluate what is gained or lost when using a lighter and more interpretable detection pipeline.

Several studies are closer to our emphasis on domain-string features and lightweight detection. The works by Koh and Rhodes [2] and by Vranken and Alizadeh [6] focus on lightweight or lexical approaches to DGA detection. Koh and Rhodes study inline DGA detection using context-sensitive word embeddings and a simple classifier, showing that lightweight models can be trained quickly and can use lexical information to classify domains. Vranken and Alizadeh examine TF-IDF-based lexical representations and compare several machine-learning models, including Logistic Regression, Random Forest, Support Vector Machines, Naive Bayes, and K-Nearest Neighbors. These studies support the central assumption behind our project: that domain strings contain useful signals for detecting DGA-generated domains.

Together, these six studies show that DGA detection has been approached through practical DNS monitoring, deep learning, feature-rich classification, and lightweight, lexical modeling. Our work fits most closely with the lightweight, lexical category, but it differs by focusing on a small set of interpretable features and comparing foundational models such as tree-based models (Random Forest). This distinction is important because a simpler model may be easier to train, interpret, and deploy, even if it does not match the full performance of heavier systems. Therefore, our study uses these six related works to frame a specific research question: how effective can lightweight Machine Learning be when DGA detection is limited to extracted lexical features from domain names alone?

3 Background

Domain Generation Algorithms are a technique that malicious actors deploy to spread malware by generating multitudes of domain names to prevent preemptive blocking. They are commonly used to aid in the dissemination of ransomware and botnets. Not all DGAs are the same, but they do have patterns in their lexical features, such as being seeded with dates, having unusual top-level-domains, or containing ran-

dom combinations of words or characters. DGAs may have a high degree of Shannon entropy (measured randomness), an unusual ratio of vowels or digits, an atypical count of subdomains and hyphens, a strange number of consonants in a row that are indicative of non-words or random consecutive numbers despite domains typically being recognizable, or an overall strange combinations of ASCII characters. They may even just be combinations of hex digits, a formulation of DWords (Double Words), embedded dates, or Punycode.

Due to the random patterns of DGAs and their ease of construction, malicious actors can utilize them to dodge block lists, but they do seem to have identifiable features. With the noticed sequences, measurements are possible to predict if a domain is intended to be for malicious purposes. Predictions may be possible by polling decision trees from Machine Learning classifiers, such as Random Forest’s majority vote.

4 Methodology

To understand the structure of DGA-generated domains, we need to train a lightweight Machine learning model on real benign and DGA domains. In this section, we discuss the methodology for designing our experiment, obtaining data sources, labeling and cleaning data, extracting features, and modeling and evaluating our findings.

4.1 Study Objective and Classification Task

We intend to figure out whether DGA-generated domains used in botnet communication can be detected using lightweight Machine Learning models based purely upon common lexical features of domain names, as opposed to relying on DNS traffic histories, WHOIS records, malware analysis, or deep learning architectures, which are often not feasible to deploy in practice.

Thus, we frame the problem as a supervised binary classification problem, where each domain is converted into a numerical feature vector and labeled as either *DGA/malicious* or *benign*. Mean differences of lexical features and Logistic Regression are used as interpretable baselines, while Random Forest is used to capture robust, nonlinear feature interactions for classification. Model performance is evaluated using accuracy, precision, recall, F1 score, ROC-AUC (Receiver Operating Characteristic - Area Under Curve), and false-positive and false-negative behavior to determine whether a lightweight, lexical pipeline is practical for DGA detection.

4.2 Data Sources and Labeling

The dataset for this study is composed of two open-source domain-name datasets. Malicious samples were obtained from the `gtkcyber/dga-dataset`, which contains 80,000 domain names associated with Domain Generation Algorithm (DGA) behavior. One million benign samples were ob-

tained from the Cisco Umbrella Top 1 Million dataset through `adebayo/cisco-umbrella-list`.

Each domain was assigned a binary class label prior to preprocessing and feature extraction. Domains from the DGA dataset were assigned the positive label (1) after removing any clearly benign domains, denoted as *DGA* or *malicious*, while domains from the Cisco Umbrella dataset were assigned the negative label (0), denoted as *benign*. However, these labels should be interpreted as dataset derived labels rather than absolute ground truth since the datasets can have mislabeled data themselves and may not contain every case of malicious or benign domains. Yet, they serve as good benchmarks for feasibility. The data was also put through lowercase transformations and had missing values eliminated to ensure consistency across all domain strings prior to feature extraction. It provides uniformity to the dataset and removes case sensitivity and probable errors.

4.3 Feature Extraction

Each domain string was standardized and converted into a numerical feature vector for model training. The extracted features included domain length, Shannon entropy, vowel ratio, digit ratio, unique-character count, hyphen count, largest number of consecutive consonants, hex ratio, possibility of an embedded Dword, largest number of consecutive digits, subdomain structure (count), top-level-domain patterns, Punycode usage, date-like substrings, the transition rate between consonants and vowels, and the character class (such as a digit, vowel, consonant, or special character) entropy.

These features were chosen because DGA-generated domains often exhibit more irregular lexical patterns than benign domains, including higher randomness, unusual character distributions, and reduced human readability. For example, an ordinary domain would not have an encrypted or shuffled date embedded into it. The resulting feature matrix was used as the input to the supervised learning models, allowing the project to evaluate whether lightweight, lexical indicators are sufficient for DGA detection.

Fig. 1 highlights the structural differences between human-readable domain names and algorithmically generated malicious ones. This figure helps push our argument forward by visually proving that DGA algorithms fail to perfectly mimic the natural character distributions of human-registered domains. It justifies our selection of these simple lexical ratios as foundational features for our lightweight Machine Learning pipeline, as they capture distinct behavioral variances between the two classes.

To better understand relationships between the extracted features a correlation matrix is shown in Fig. 2. Features that have higher correlations with one another will typically overlap in Random Forest analyses and have less decision making power as the correlations mean they help detect similar patterns in DGA domains. Other features correlate with labels,

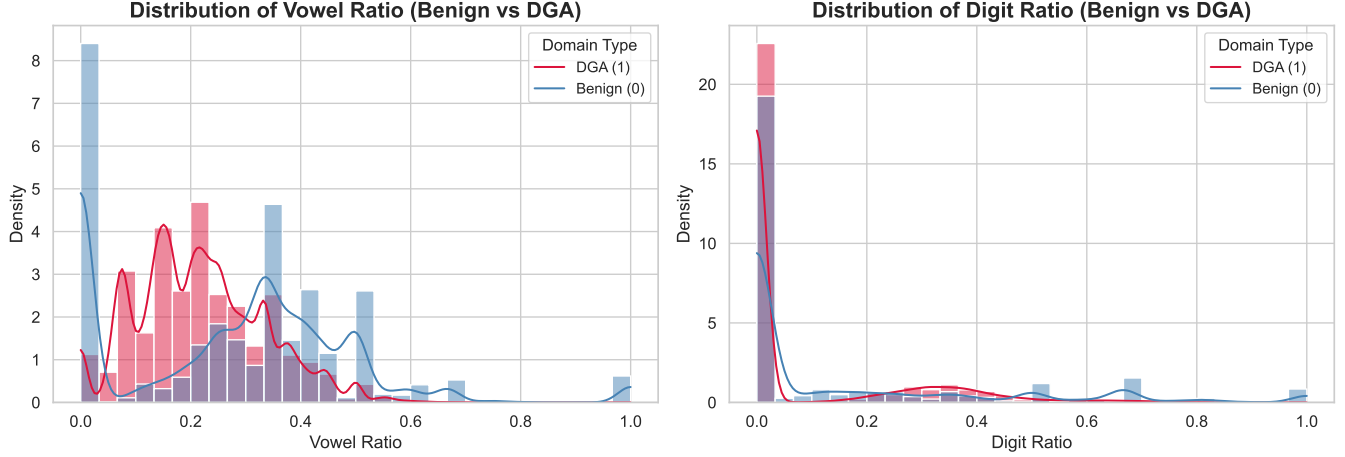


Figure 1: **DGA Domains Exhibit Lower Vowel Ratios While Digit Ratios Heavily Overlap with Benign Samples** — Represented is the distribution of vowel ratio (left) and digit ratio (right) across the dataset. DGA domains tend to have fewer vowels compared to other characters in a domain string, whereas benign domains tend to contain more vowels or none at all. Both classes exhibit a high degree of overlap in their digit ratio distributions, making single-feature separation insufficient.

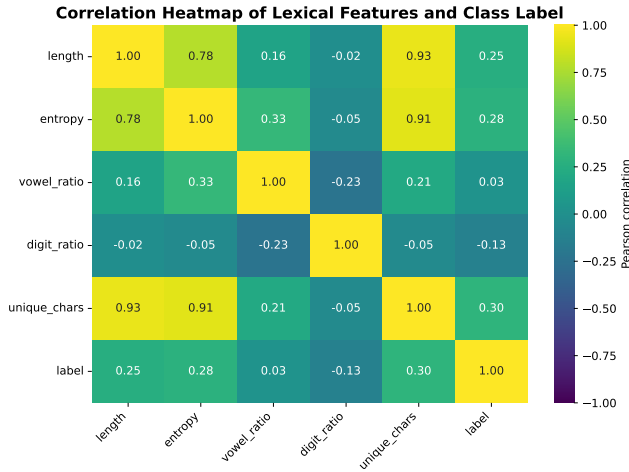


Figure 2: **Lexical Features Show Strong Mutual Correlation but Low Direct Linear Correlation to Class Labels** — Portrayed is the correlation heatmap of the extracted lexical features and the binary class label. Certain features correlate heavily with one another, providing redundant information, while direct feature-to-label linear correlations remain low overall, justifying the need for a non-linear classifier.

representing a possibility that they can help depict benign and DGA domains. However, these correlations are low, proving that singular feature separation is not enough.

4.4 Modeling Procedure

After feature extraction, each domain was represented as a feature vector X with a corresponding binary label y , where $y = 0$ indicates a benign domain and $y = 1$ indicates a DGA (malicious) domain. The dataset was split into training and testing sets using a fixed random state to ensure reproducibil-

ity.

Random Forest was our main focus for determining the feasibility of our lightweight model between a balanced dataset of 80,000 benign and DGA domains and one that focused on mimicking real-world DNS traffic with the DGA and benign domain split being 7.4% and 92.6%, respectively, for 1080000 samples. To note, benign-biased data better reflects DNS traffic. The Random Forest classifier was used to combine all the features together and make decisions based on the classification from multiple features simultaneously. The Random Forest classifier provides a deeper inspection of the features working together for classification. Hyperparameter tuning was performed on the training set, and the final models were evaluated on the held-out test set to measure generalization performance.

4.5 Evaluation and Analysis

Model performance was evaluated using accuracy, precision, recall, F1 score, and false-positive behavior. Accuracy measures the overall proportion of correct classifications, while precision and recall provide more detail about the quality of DGA detection. Precision measures how often domains predicted as DGA were actually malicious, and recall measures how many true DGA domains were successfully detected. The F1 score was used as a balanced metric between precision and recall, which is important because high accuracy alone may hide poor detection behavior in a class.

The Random Forest model was also analyzed across different numbers of trees to determine how the complexity of the model affected the performance. Based on the current results, the Random Forest achieved very high classification performance, with the best reported accuracy, precision, recall, and F1 score being 1.0. The best-performing configuration

used a small number of trees for most metrics, suggesting that the extracted lexical features provide strong separation between benign and DGA domains without requiring a large or computationally expensive ensemble.

In addition to classification metrics, feature-level analysis was used to interpret why the model performed well among other tests. Correlation heatmaps were used to examine relationships between features and the binary class label, while Random Forest feature importance was used to identify which features contributed most strongly to classification. The current feature-importance results indicate that suspicious top-level-domain patterns, subdomain count, long consonant sequences, hyphen count, and domain length were among the strongest predictors. This analysis helps connect the model’s decisions back to interpretable lexical properties rather than treating the classifier as a black box.

False positives are especially important for this project because incorrectly flagging benign domains could disrupt normal DNS traffic in a real deployment. Therefore, the final analysis will not rely only on aggregate accuracy. It will examine false-positive and false-negative rates to determine whether the lightweight lexical pipeline is practical for distinguishing DGA-generated malicious domains from benign domains in a realistic security setting.

5 Results

After collecting the data from the balanced model, in which the length of the benign domain dataset was equivalent to the length of the DGA domain dataset, we were able to make multiple conclusions about the model efficacy being promising. As per Fig. 3, we have 7 unique DGA families within our dataset, all presenting differing techniques for generating malicious domains. The domain lengths between the DGA families vary, having distinct patterns supporting the idea of lexical features for DGA classification and anomaly detection as per Fig. 4.

Furthermore, based on mean differences between benign and DGA domains, the most important features for distinguishing between the classes are domain length at 8.607, unique characters at 5.986, the longest set of consecutive consonants at 3.802, and subdomain count at -2.267. This means that DGA domains typically are 8.6 characters longer, have almost 6 more unique characters, have 4 more consonants in a row, and have 2.2 less subdomains than benign domains.

To account for overlapping features during analysis, we conducted a Random Forest classifier as it can combine all the features together and make decisions based on the classification from multiple features simultaneously. The Random Forest classifier provides a deeper inspection of the features working together for classification. Fig. 5 represents the relative feature importance as projected by the Random Forest classifier optimized on the F1 score. Utilizing a balance measure of a model’s accuracy based on the harmonic mean of

precision and recall, it is clear that the input variables that contribute the most to the model’s predictive power are the subdomain count at 50.6%, the largest number of consecutive consonants in the domain at 22.4%, malicious top-level domains at 18.0%, unique characters at 4.7%, and the number of hyphens in the domain at 2.5%. Most other features provide extremely little information to make decisions about whether a domain is benign or malicious. More noted features, such as Shannon entropy, domain length, vowel ratio, and digit ratio, contribute to less than 2% of the predictive power in our lexical feature model. This suggests that the structure of the domain is more discriminative than individual character-level randomness. As for our argument, this figure highlights a methodological concern for subdomain count due to its overwhelming decision-making power, but more importantly represents the effectiveness of our model’s novel lexical features beyond simple baseline features. The model is proven to be lightweight as well since the number of trees to achieve a perfect metric of scoring 100% for accuracy, precision, recall, and F1 score are 2, 2, 1, and 2 respectively. However, it is important to note that 100% is extremely concerning for data leakage. To further this skepticism, the false positive rate and false negative rates for all scores are 0%, excluding recall which individually has a false positive rate of 0.019%. Misclassifications of DGA domains may be due to lexical properties that more closely resemble that of a benign domain, such as more human-readable patterns, more subdomains, or shorter domain strings, while the misclassifications for benign domains may be from the inverse. As a final note, Logistic Regression provided very little information for accuracy, precision, recall, and F1 score as it also provided a 99.9% to 100% score for all metrics. The skewed feature distribution suggests that a model utilizing subdomain count, consecutive consonants, and top-level domains could be a minimal, deployable configuration to achieve full model performance.

To test our concerns, we removed the top-level domain from the dataset as it is the most concerning feature. It was trained on the entire dataset and is a criticized feature due to it being trivially easy for malware authors to change according to Koh and Rhodes [2], so it was removed from the analysis. In this case, the best accuracy, precision, and F1 score were 99.9% while recall was 100%, requiring polling from 11 decision trees to make a decision (although fewer trees for similar results are also available with this data set to provide for a faster, live analysis of domains). In this case according to the F1 score, subdomain count has a decision making power of 47.7%, domain length with 14.8%, maximum consecutive consonants at 13.4%, unique character appearances has 8.2%, hyphen count with 4.6%, and Shannon entropy at 4.6%. In this case, all false negative rates were 0% while all false positive rates were 0.006%. Subdomain count still controls most of the decision making power but with a slightly higher emphasis on more typical features such as entropy and length.

DGA Domain Families

Family	Count	Example 1	Example 2	Example 3
cryptolocker	37254	glbtlxwwhbnpxs	xxmamopyipbfpk	jpqftymiuver
newgoz	9276	zfd5szpi18l85wj9uy13l69rg	mzapiwbcsbqlyxnla1hprx8n	n3m4txqb6a415iwzkrmdzgk6
gameoverdga	8461	6xzxsw3sokvg1tc752y1a6p0af	1rw1pc44mkxp4qjx18qua26	1x75xo41t4fno51wc86e1y4s13d
nivdort	8456	takenroll	dreamdont	visitshot
necurs	8331	sfmbdpnijlarq	vdqqfcdax	njtvrfhfakiho
goz	6136	rcdqeqmpxgdiuqombhuwcfu	jrxrovxbemgqutopblbbfe	dixnzhelzukunftuivlamadu
bamital	2086	5df9d44072820ad6eaac91669f357fea	a7fd378e8e64200f13421e6135a06dce	dd82f8a0b2b6520e2706f704ba2a289a

Figure 3: **CryptoLocker and newGOZ Dominate the Malicious Sample Family Distribution** — The table is a representation of DGA domain families within the project dataset. CryptoLocker is the most prominent family, relying on pseudo-random numbers seeded with UTC dates, followed by newGOZ, which generates domains using a hardcoded mathematical formula and cryptographic hashes.

Computing with a Stratified K-Fold with 5 splits as a cross-validation technique, we found the best feature combinations containing 3 to 8 features with and without top-level domains being analyzed. The best combination was with malicious top-level domain, subdomain count, vowel ratio, unique characters, and embedded dates, which achieved an accuracy, precision, recall, and F1 score of 100% with only one tree. The subdomain count is credited with 99.9% of the decision making power for this model and malicious top-level domains accounts for 0.01%. The false positive and false negative rates are 0 in all categories. Without malicious top-level domains, the number of decision trees needing to be polled rises to 7, although fewer are possible for similar results. A sample tree can be found at Fig. 6. The best accuracy, precision, and F1 score is 99.9% and recall is 100%, where the combination is approximately subdomain count at 57.1% decision making power, unique characters at 15.3%, domain length at 13.1%, hyphen count at 9.3%, vowel ratio at 3.6%, digit ratio at 1.0%, maximum consecutive digit count being 0.3%, and hex ratio at 0.2%. The false positive rate for all categories was 0.006% and false negative rate is 0%. Although we notice a shifting in important feature combinations, subdomain still seems to be dominating the decision making. Feature ablation reveals a dependence on structural characteristics, specifically subdomain count in this instance.

Due to the concerns of over relying on malicious top-level domains and subdomain counts, another dataset excluding these was conducted and provided promising results to the capabilities of our model. In this case, the metrics differ by a decent margin. Best accuracy scores at 92.4% with 65 trees, F1 score at 92.5% with 65 trees, recall at 94.5% with 14 trees, and precision at 90.6% with 10 trees. As usual, the number of trees can be lowered and achieve similar results to the best metrics, decreasing the computational cost. This combination has F1 score feature importance values of domain length at 44.1% decision power, maximum consecutive consonants at 35.0%, hyphen count at 14.4%, digit ratio at 4.2%, maximum consecutive digits at 2.2%, and Punycode at just above 0%. The false positive rate for each metric is between 9.7% to

10.3%, while false negative rate is between 5.1% to 5.6%, a higher rate than the previous examples. Lastly, the ROC-AUC curve for the first time is not 100% for the entire separation as per Fig. 7. Instead of perfect separation, we are achieving excellent separation of classes at 98.4%, which is less likely from data leakage or another unforeseen data element. Although including subdomain count and malicious top-level domains is not incorrect, this case provides an observable reality of a less optimistic approach to DGA detection accuracy and aligns with more commonly studied features for DGA domain detection. Furthermore, the Logistic Regression has dropped to approximately 90% after consistently scoring 99.9% to 100% in the accuracy, precision, recall, and F1 score metrics, meaning that a linear model may also be possible in this dataset. However, the Random Forest classifier does seem to account for nonlinear interactions, achieving better performances in trade for the speed acquired from composing a linear model. Lexical features that do not rely on structural dominance still prove by our tested metrics to be a valuable way of separating DGA and benign domains.

When looking at the other classification model in which the split between the DGA and benign domains is 7.4% and 92.6%, a dataset leaning towards benign dominance since it more closely resembles real-world DNS traffic, most cases barely differ. The feature importance may change by approximately +10% for the most important feature lists and some of the combinations may vary, but the overall conclusions are the same. It is important to note that the false positive and negative rates did increase, but they are still below 1% in these cases; thus, they do not reasonably merit any mention for their specific variations. However, the best metric percentages varied significantly when removing both the top-level domain and subdomain count features, where the best accuracy score with 10 trees was 97.6%, precision with 46 trees was 90.8%, recall with 3 trees was 75.5%, and F1 score with 3 trees was merely 81.8%. In this case, the false negative rate rises to between 24% and 26% and false positive rate is about 0.6% and to 0.7% for all metrics. Additionally, Logistic Regression significantly decreased in most metrics,

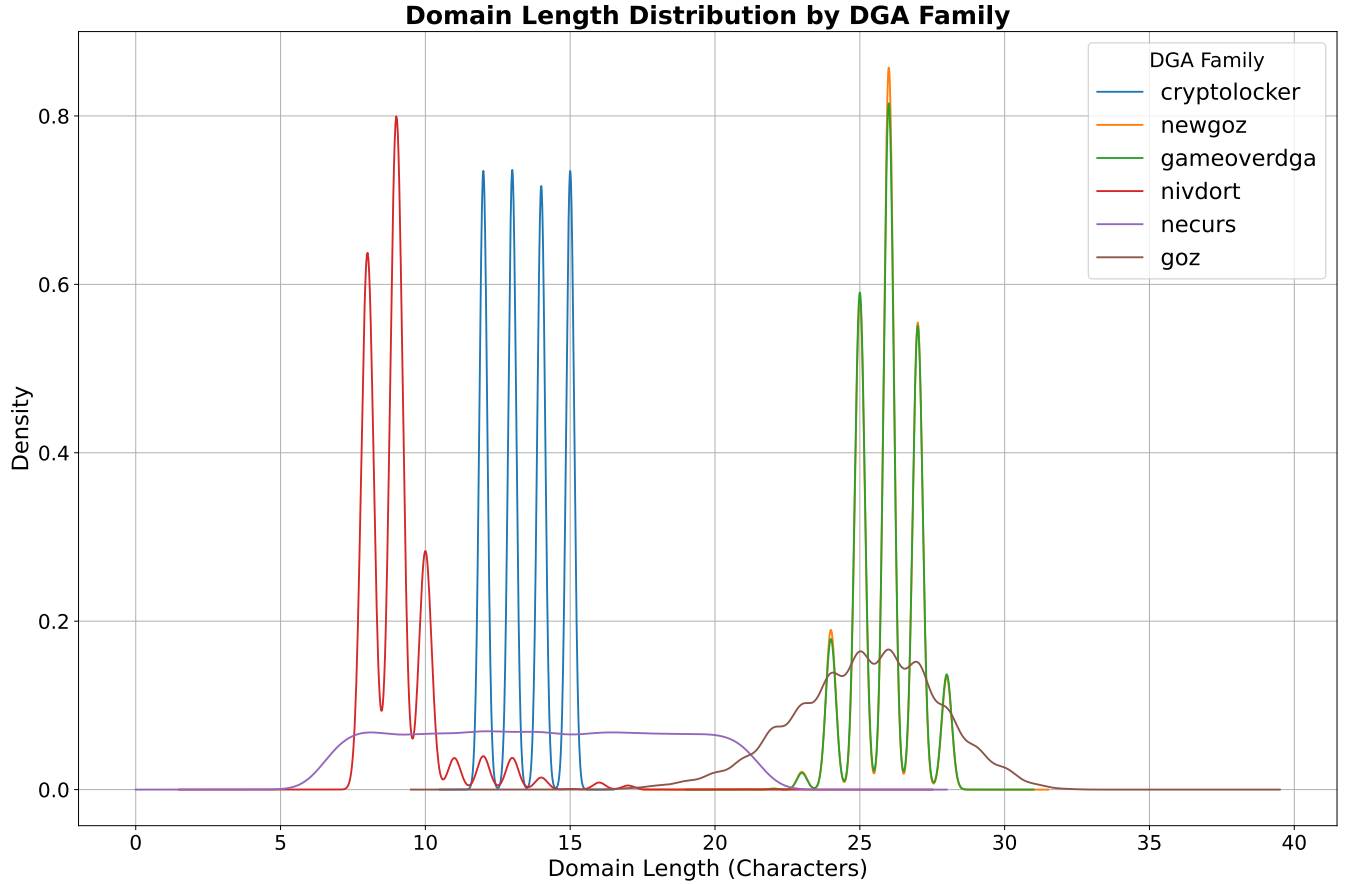


Figure 4: **Distinct DGA Families Produce Highly Identifiable Domain Length Footprints** — Domain length distribution curves grouped by specific DGA family are visualized. For instance, CryptoLocker families produce a narrow distribution concentrated between 12 to 15 characters, whereas GOZ features a wider, bell-shaped distribution ranging from 16 to 31 characters.

scoring 96.4% in accuracy, 82.7% in precision, 63.8% in recall, and 72.0% in F1 score. A model like this may be trivially easy for a sophisticated hacker to manipulate and dodge detection as the model is structurally dependent. The present DGAs may be better disguised to mimic real words, and if solely dependent on structural features, a malware author can avoid suspicious subdomain lengths and top-level domains to disguise themselves further. Moreover, with such a high false negative rate, this model would not be deployable as a standalone DGA detector. The found discrepancy between training on the classification of the real-world model illustrates some defining information about the lack of practical deployability of a model in a different context compared to the chosen balanced dataset (50/50).

As a last note, while previous deep learning models for DGA detection (e.g., LSTMs) are computationally expensive and difficult to deploy in real-time, our approach effectively bridges deep learning with lightweight, lexical classification. By utilizing a pre-trained GPT-2 model, we extracted language perplexity (`llm_perplexity`) as a single, quantifiable feature to evaluate a domain’s structural unnaturalness.

Our analysis shows a solid positive correlation of 0.24 between this LLM-derived feature and the DGA label. This confirms that distilling an LLM’s language understanding into a lightweight feature successfully captures the non-human logic of DGA strings without sacrificing the operational efficiency of our Random Forest model.

The lightweight, lexical Machine Learning model effectively separates benign and DGA domains. It achieves this in most cases with and without utilizing structural features of the domain name string, finding multiple critical signal combinations for defining DGA behavior. The model is deployable through real-world performance metrics and constantly scores highly on accuracy, precision, recall, F1 score, and ROC-AUC, while having low false positive and false negative behaviors.

6 Discussion

The results show that lightweight lexical features can be useful for detecting DGA-generated domains, but they also show that not all features contribute equally. The domain-family analysis in Fig. 3 and Kernel Density Estimation of domain

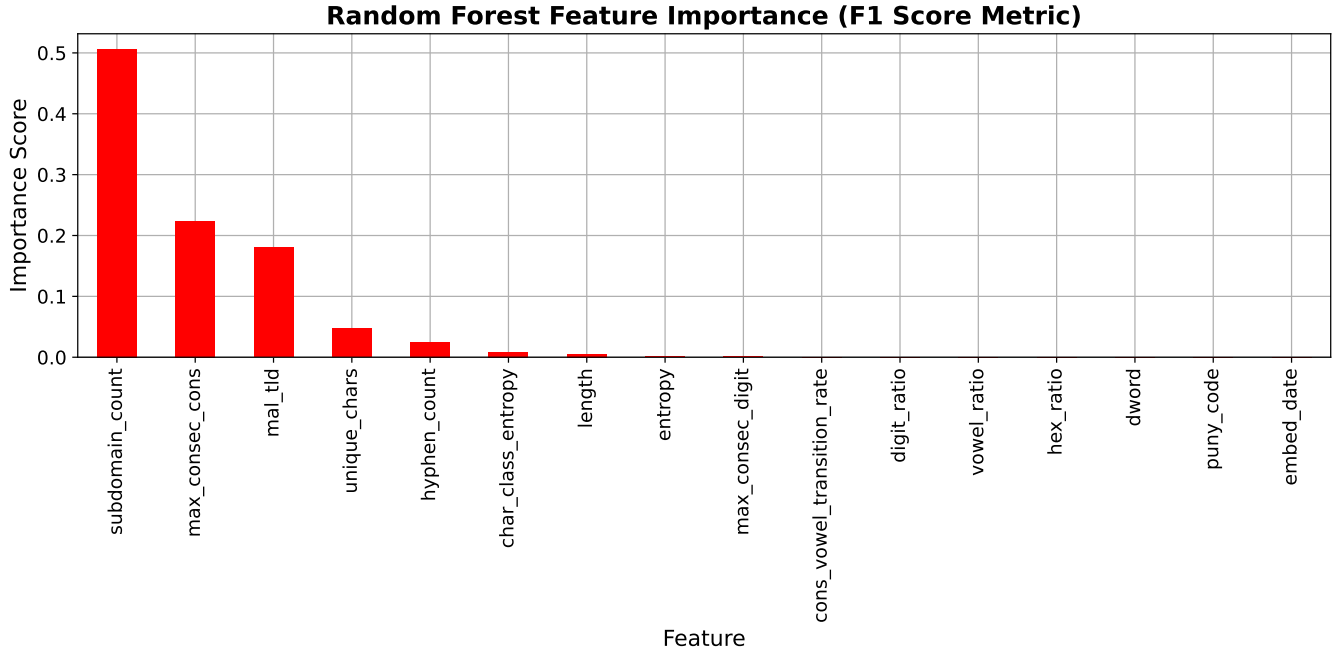


Figure 5: **Structural Properties and Readability Metrics Drive Over 90% of Model Decisions** — A Random Forest feature importance optimized on the F1-score metric is composed. Subdomain count, maximum consecutive consonants, and malicious TLD patterns heavily dominate the classification process, proving that domain structure is far more discriminative than simple randomness metrics like Shannon entropy.

length distribution by DGA family in Fig. 4 support the idea that different DGA families produce distinct lexical patterns. For example, CryptoLocker and newGOZ appear frequently in the dataset, and their domain length distributions are noticeably different. This suggests that even simple features, such as length and character structure, can capture meaningful differences between benign and malicious domains.

The Random Forest feature importance results provide the clearest insight into how the model makes decisions. The strongest features were subdomain count, maximum consecutive consonants, suspicious top-level domain, unique-character count, and hyphen count. Together, subdomain count, consecutive consonants, and top-level-domain features accounted for over 90% of the model’s decision-making. This shows that the model is not mainly relying on general randomness features like Shannon entropy, vowel ratio, or digit ratio. Instead, domain structure and readability patterns appear to be more useful for this dataset.

However, this also reveals a limitation. The model’s very high performance may partly come from structural patterns in the dataset rather than from fully general DGA behavior. Subdomain count and suspicious top-level-domains are powerful features, but they do not always generalize well. A malware author could switch to more common top-level-domains or imitate benign subdomain structures. Because of this, the near-perfect Random Forest results should be interpreted with caution rather than treated as proof that the model is deployment-ready.

The ablation results support this concern. When suspicious top-level-domain features were removed, the model still performed extremely well, but subdomain count remained dominant. When both top-level-domain and subdomain-count features were removed, performance dropped to a more realistic level. In that case, features like domain length, maximum consecutive consonants, hyphen count, digit ratio, and maximum consecutive digits became more important, complementing other research that came to similar conclusions. This suggests that the model still has useful lexical detection ability, but its strongest results depend heavily on a small number of structural indicators.

The realistic class-imbalance experiment also shows that accuracy alone is not enough to evaluate deployability. In real DNS traffic, benign domains are far more common than DGA domains. Under this setting, the model still achieved high accuracy, but recall dropped when the strongest structural features were removed. This means the model may miss more malicious domains in practical use, even if the overall accuracy looks strong.

Overall, the model is promising as a lightweight first-stage detector. Compared with deep learning systems, it is easier to train, faster to run, and more interpretable. Our model does hold up according to our metrics when compared to similar lexical based models. However, it should not be used as a standalone blocking system without additional validation. The best practical use would be as a triage layer that flags suspicious domains for further analysis using DNS behavior,

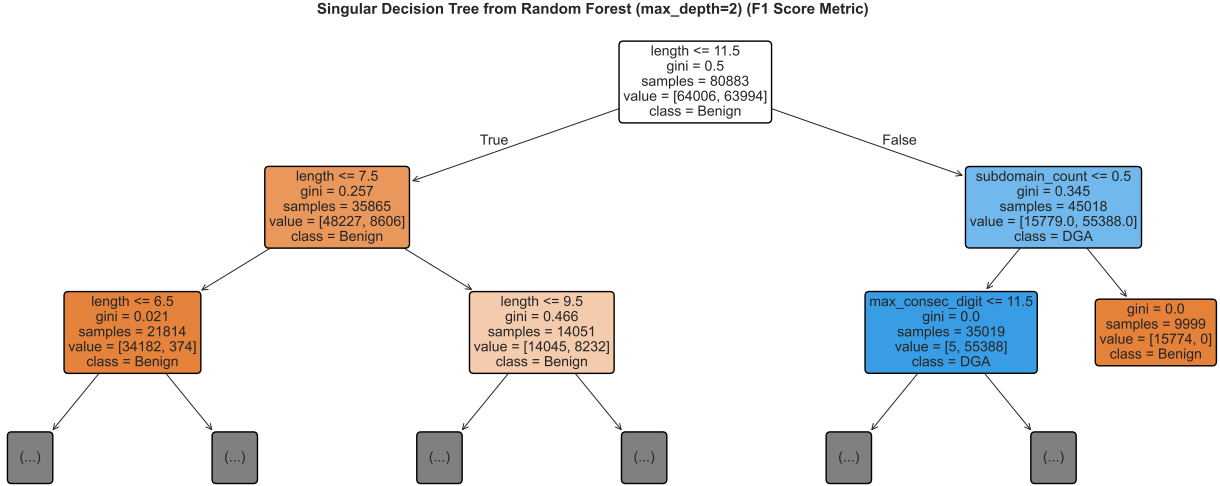


Figure 6: **Ensemble Voting Resolves Complex Lexical Logic via Individual Component Trees** — A singular decision tree extracted from the Random Forest model trained without malicious TLD features is shown. This tree illustrates the sequential, conditional classification path for a domain string, which subsequently contributes to the final majority poll.

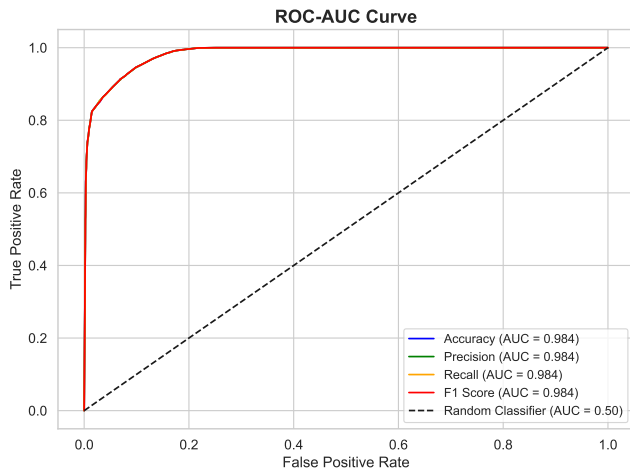


Figure 7: **Lexical Pipeline Retains Strong Separation Without Structural Features** — The ROC-AUC curve yields an AUC of 0.984, demonstrating excellent classification performance.

reputation data, or analyst review. Future work should test the model on more DGA families that may have not appeared in this dataset, newer and alternative datasets, live DNS traffic, and adversarial domains designed to mimic benign lexical patterns. A study can also delve into the features that make the lighter-weight Logistic Regression a viable model as our research has shown some viable deployability of such work, despite it scoring less in metrics when compared to the Random Forest model.

7 Ethical Considerations

This project uses publicly available domain-name datasets, so it does not involve collecting private user data or interacting with live malicious infrastructure. Since the model only analyzes domain strings, the direct privacy risk is low.

The main ethical concern is that this research is dual-use. By identifying which lexical patterns make DGA domains easier to detect, we may also reveal ways for attackers to avoid detection. For example, malware authors could generate domains with more natural lengths, fewer long consonant sequences, or more common top-level domains.

False positives are another concern. If a benign domain is incorrectly flagged as malicious, it could block normal user traffic or disrupt legitimate websites. False negatives are also harmful because missed DGA domains may allow botnet communications to continue. Because of this, the model should not be used as the only basis for automatic blocking. Overall, this model is best used as a lightweight detection aid alongside other security tools and analyst reviews, while continuing to be retrained on newer datasets.

8 Conclusion

Lightweight Machine Learning techniques can effectively detect many DGA-generated domains using only lexical features from the domain name. The Random Forest model performed well, showing that simple domain-string features can still provide strong classification power without requiring DNS traffic history, WHOIS data, or deep learning models. The most useful features were subdomain count, maximum consecutive consonants, suspicious top-level-domain patterns,

unique-character count, hyphen count, and domain length. These results suggest that DGA domains are often distinguishable not just because they are “random,” but because they have different structure and readability compared to benign domains. The project also showed important limitations. When top-level-domain and subdomain-count features were removed, model performance dropped, meaning the strongest results partly depend on features that may not always generalize. In the more realistic class-imbalance test, recall also decreased, showing that real-world deployment training is harder than balanced testing may suggest. Therefore, this model should be treated as a lightweight first-stage detector, not a complete standalone security system. Compared with heavier deep learning approaches, our model is faster, simpler, and easier to interpret. However, it may be easier for attackers to evade if they design DGA domains to look more like normal domains. Future work should test the model on more DGA families, live DNS traffic, newer datasets, and adversarial examples. Overall, the main takeaway is that lightweight lexical detection is highly promising, but larger-scale validation on more diverse and realistic DNS data is still needed before it can be confidently deployed in real-world security systems.

Authorship & Contribution Statement

Sean Silver: Random Forest Classifier, Logistic Regression, Data Collection and Cleaning, Lexical Feature Analysis, ROC-AUC, Scoring Metrics, Decision Trees, True Positive, True Negative, False Positive, and False Negative Rates, DGA Domain Families, Domain Length Distributions, Writer, and Reviewer

Ethan Ge: LLM Perplexity, Feature Statistics, Data Collection and Cleaning, Lexical Feature Analysis, Writer, and Reviewer

Lihao Pan: Heatmap Correlations, Vowel and Digit Ratio Distributions, Writer, and Reviewer

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